Elevator pitch 🎯

This repo is a small production-style hybrid search pipeline for PDFs/DOCX: it ingests files, extracts text (OCR fallback), deduplicates, token-chunks documents, precomputes multilingual embeddings, builds FAISS ANN indexes, and serves a hybrid search that combines BM25 (OpenSearch) + ANN (FAISS) to rank whole documents with snippet-level evidence.

One-minute explanation (what to say first in an interview) 🕒

- "We built a hybrid retrieval pipeline where BM25 provides precise lexical matches and FAISS (IVF+PQ + optional HNSW hot tier) provides semantic recall. Documents are normalized, chunked to ~400–800 tokens, embedded with a multilingual MiniLM, then indexed. At query time we run BM25 and ANN in parallel, union chunk candidates, aggregate per-doc signals (max BM25, max vector), and return ranked documents with a snippet and page offset."

System components — 30-second walk-through 🔧

- **Ingest (**optimized_ingest.py**)**: PDF/DOCX extraction, Tesseract OCR fallback, per-page OCR confidence, language detection (fastText + Lingua).
- **Dedup (**dedupe_simhash.py**)**: document-level Simhash dedupe to avoid redundant indexing.
- **Chunking (**chunker.py**)**: tokenizer-based chunking with overlap (target tokens ~512, overlap ~64).
- **Embeddings (**embedder.py**)**: batch embed with sentence-transformers; normalize embeddings and write .npy + meta.
- **ANN build (**build_faiss.py**)**: create IVF+PQ (and HNSW for hot subsets) from embeddings.
- **Search (**search_service.py**)**: OpenSearch BM25 + FAISS ANN in parallel, union & aggregate to score docs.